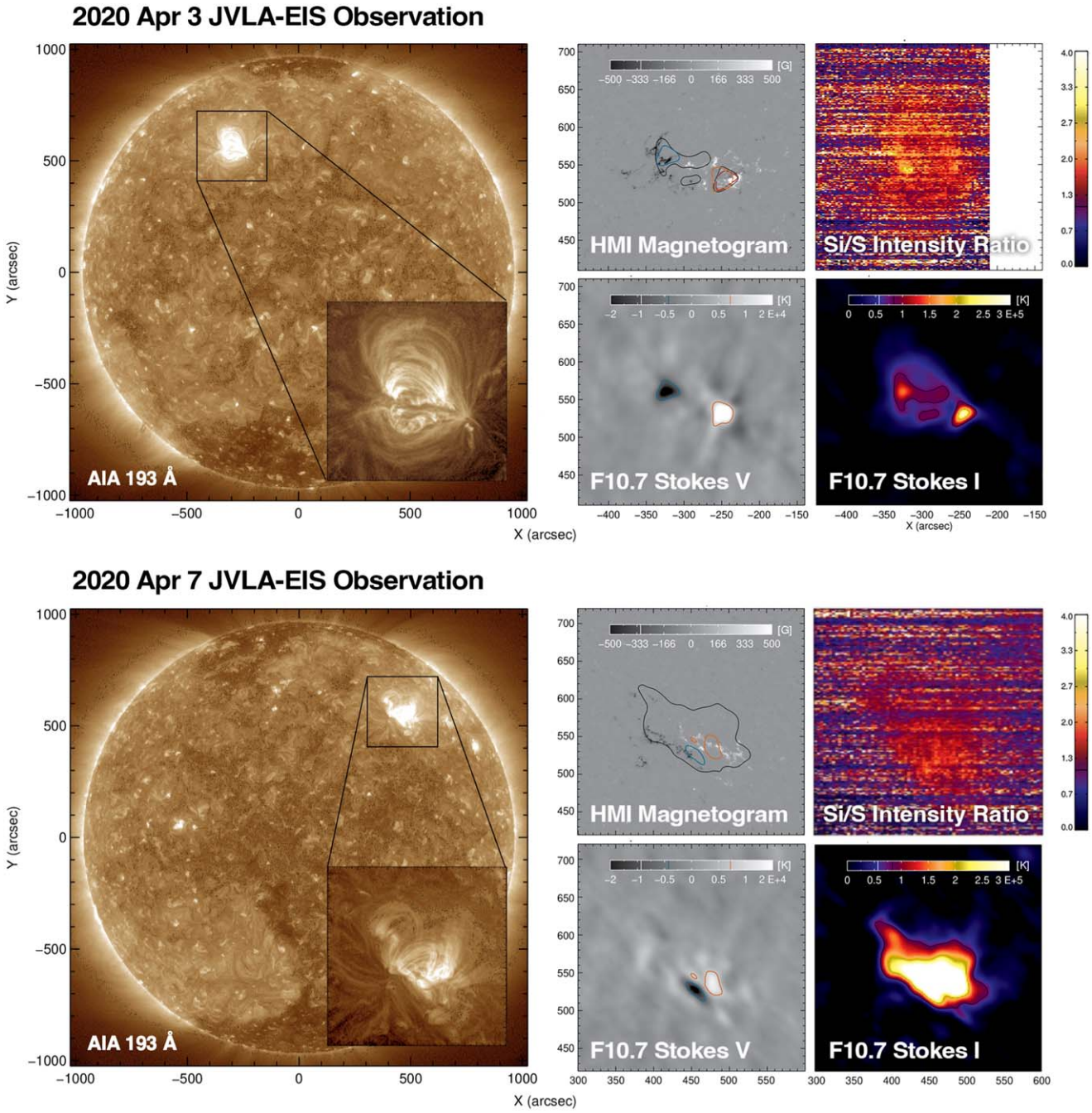

Figure 1. Small bipolar active region AR 12759, observed during the JVLA/20A-047 observing campaign on April 3 and 7. Left to right, top to bottom: AIA 193 Å, HMI magnetogram, Si X 258.38 Å/S X 264.23 Å intensity ratio map, F10.7 radio flux map (Stokes V), and F10.7 radio flux map (Stokes I). Four F10.7 cm regions are used in this paper to calculate the FIP bias: (1) a Stokes I region with a brightness temperature >80,000 K subtracted by strong Stokes V emission (black contour), (2) negative Stokes V regions with a brightness temperature <−10,000 K (blue contour), (3) a positive Stokes V region with brightness temperature >10,000 K (orange contour), and (4) the estimated gyroresonance region on April 3 defined using ((Stokes I-modeled free–free)/Stokes I) (red dashed contour; Section 4.1). It can be seen that the negative Stokes V region is associated with the following polarity, whereas the positive Stokes V and gyroresonance regions are associated with the leading polarity. The Si/S intensity ratio maps shown here are for demonstration purposes. In our analysis, we calculated and used the spatially averaged FIP bias. This significantly improves the signal-to-noise ratio.

to minimize the instrumental offset between the three instruments. First, the AIA coordinate system was used as our base coordinate system. Second, as the active region was stable throughout the EIS raster duration, the Fe XII 195.12 Å intensity maps observed by EIS were aligned with an AIA 193 Å image taken at the beginning raster time. Lastly, to align JVLA to EIS and AIA, we followed the approach of White et al. (2000) and Schonfeld et al. (2015) to estimate the free–free component of the F10.7 flux using AIA DEM (Hannah & Kontar 2012). This allowed us to visualize F10.7 flux data in AIA coordinates. The F10.7 cm bremsstrahlung emission taken by JVLA was then aligned to this predicted emission. As a final check, the coronal magnetic field of AR 12759 was modeled by extrapolating radial field magnetograms that were taken by HMI on April 3 and 7 (see Section 2.3). The F10.7 cm gyroresonance observations at 2.8 GHz originate from thin